SQL 03: SELEKSI BARIS DENGAN WHERE

Demo Query | Query #1 | Query #2 | Upgrade

```
show databases;
use classicmodels;
show tables;
describe employees;

select * from employees;
select `lastName`, `firstName`, `jobTitle`
from employees;
```

Auto Commit | Manual | Save | Run Selection

	lastName	firstName	jobTitle
1	Murphy	Diane	President
2	Patterson	Mary	VP Sales
3	Firrelli	Jeff	VP Marketing
4	Patterson	William	Sales Manager (APAC)
5	Bondur	Gerard	Sale Manager (EMEA)
6	Bow	Anthony	Sales Manager (NA)
7	Jennings	Leslie	Sales Rep
8	Thompson	Leslie	Sales Rep
9	Firrelli	Julie	Sales Rep
10	Patterson	Steve	Sales Rep
11	Tseng	Foon Yue	Sales Rep

23 | 0 affected | 10ms | Download

Dari banyaknya jobtitle di atas, kita dapat memfilter hanya untuk pekerjaan tertentu saja yang ingin kita tampilkan caranya dengan menggunakan klausa WHERE. Berikut adalah hasilnya:

Demo Query | Query #1 | Query #2 | Upgrade

```
show databases;
use classicmodels;
show tables;
describe employees;

select * from employees;
select `lastName`, `firstName`, `jobTitle`
from employees
where `jobTitle` = 'Sales Rep';
```

Auto Commit | Manual | Save | Run Selection

	lastName	firstName	jobTitle
1	Jennings	Leslie	Sales Rep
2	Thompson	Leslie	Sales Rep
3	Firrelli	Julie	Sales Rep
4	Patterson	Steve	Sales Rep
5	Tseng	Foon Yue	Sales Rep
6	Vanauf	George	Sales Rep
7	Bondur	Loui	Sales Rep
8	Hernandez	Gerard	Sales Rep

17 | 0 affected | 17ms | Download

Lalu, ada kebutuhan untuk menambahkan kolom officeCode:

Demo Query | Query #1 | Query #2 | Upgrade

```
show tables;
describe employees;

select * from employees;
select `lastName`, `firstName`, `jobTitle`, `officeCode`
from employees
where `jobTitle` = 'Sales Rep';
```

Auto Commit | Manual | Save | Run Selection

	lastName	firstName	jobTitle	officeCode
1	Jennings	Leslie	Sales Rep	1
2	Thompson	Leslie	Sales Rep	1
3	Firrelli	Julie	Sales Rep	2
4	Patterson	Steve	Sales Rep	2
5	Tseng	Foon Yue	Sales Rep	3
6	Vanauf	George	Sales Rep	3
7	Bondur	Loui	Sales Rep	4
8	Hernandez	Gerard	Sales Rep	4
9	Castillo	Pamela	Sales Rep	4
10	Bott	Larry	Sales Rep	7
11	Jones	Barry	Sales Rep	7

17 | 0 affected | 18ms | Download

Kemudian, ada 2 kondisi kita ingin buat jobtitle nya adalah Sales Rep dan officeCode nya hanya 1, maka kita dapat gunakan operator logika AND:

Demo Query | Query #1 | Query #2 | Upgrade

```
show tables;
describe employees;

select * from employees;
select `lastName`, `firstName`, `jobTitle`, `officeCode`
from employees
where `jobTitle` = 'Sales Rep' AND `officeCode` = 1;
```

Auto Commit | Manual | Save | Run Selection

	lastName	firstName	jobTitle	officeCode
1	Jennings	Leslie	Sales Rep	1
2	Thompson	Leslie	Sales Rep	1

Selanjutnya ada operator logika OR untuk menampilkan salah satu data yang benar/TRUE:

Demo Query | Query #1 | Query #2 | Upgrade

```
show tables;
describe employees;

select * from employees;
select `lastName`, `firstName`, `jobTitle`, `officeCode`
from employees
where `jobTitle` = 'Sales Rep' OR `officeCode` = 1;
```

Auto Commit | Manual | Save | Run Selection

	lastName	firstName	jobTitle	officeCode
1	Murphy	Diane	President	1
2	Patterson	Mary	VP Sales	1
3	Firrelli	Jeff	VP Mark...	1
4	Bow	Anthony	Sales Ma...	1
5	Jennings	Leslie	Sales Rep	1
6	Thompson	Leslie	Sales Rep	1
7	Firrelli	Julie	Sales Rep	2
8	Patterson	Steve	Sales Rep	2
9	Tseng	Foon Yue	Sales Rep	3
10	Vanauf	George	Sales Rep	3
11	Bondur	Loui	Sales Rep	4

21 | 0 affected | 14ms | Download

Kita juga bisa menambahkan klausa ORDER BY untuk mengurutkan kolom misalnya berdasarkan officeCode dan jobTitle:

```
select * from employees;
select `lastName`, `firstName`, `jobTitle`, `officeCode`
from employees
where `jobTitle` = 'Sales Rep' OR `officeCode` = 1
order by `officeCode`, `jobTitle`;
```

Auto Commit | Manual | Save | Run Selection

	lastName	firstName	jobTitle	officeCode
1	Murphy	Diane	President	1
2	Bow	Anthony	Sales Ma...	1
3	Thompson	Leslie	Sales Rep	1
4	Jennings	Leslie	Sales Rep	1
5	Firrelli	Jeff	VP Mark...	1
6	Patterson	Mary	VP Sales	1
7	Firrelli	Julie	Sales Rep	2
8	Patterson	Steve	Sales Rep	2
9	Tseng	Foon Yue	Sales Rep	3
10	Vanauf	George	Sales Rep	3
11	Castillo	Pamela	Sales Rep	4

21 | 0 affected | 18ms | Download

Kita juga bisa mengakomodir kebutuhan untuk rentang officeCode misalnya 1-3 saja:

```
select * from employees;
select `lastName`, `firstName`, `jobTitle`, `officeCode`
from employees
where `officeCode` between 1 and 3
order by `officeCode`, `jobTitle`;
```

Auto Commit Manual

	lastName	firstName	jobTitle	officeCode
1	Murphy	Diane	President	1
2	Bow	Anthony	Sales Ma...	1
3	Thompson	Leslie	Sales Rep	1
4	Jennings	Leslie	Sales Rep	1
5	Firrelli	Jeff	VP Mark...	1
6	Patterson	Mary	VP Sales	1
7	Firrelli	Julie	Sales Rep	2
8	Patterson	Steve	Sales Rep	2
9	Tseng	Foon Yue	Sales Rep	3
10	Vanauf	George	Sales Rep	3

Lalu kita bisa memodifikasi bagian lastname hanya nama yang berakhiran son saja yang kita tampilkan, caranya dengan menggunakan operator LIKE dan penambahaan karakter %, % merepresentasikan untuk karakter apapun juga dan berapapun juga.

Contoh:

```
select * from employees;
select `lastName`, `firstName`, `jobTitle`, `officeCode`
from employees
where `lastName` like '%son'
order by `officeCode`, `jobTitle`;
```

Auto Commit Manual

	lastName	firstName	jobTitle	officeCode
1	Thompson	Leslie	Sales Rep	1
2	Patterson	Mary	VP Sales	1
3	Patterson	Steve	Sales Rep	2
4	Patterson	William	Sales Ma...	6

Adapun karakter underscore _ merepresentasikan untuk karakter apapun dengan jumlah 1. Misalnya kita ingin menampilkan firstname dengan nama pada huruf keduanya adalah hanya a saja, maka:

```sql
select `lastName`, `firstName`, `jobTitle`, `officeCode`
from employees
where `firstName` like '_a%'
order by `officeCode`, `jobTitle`;
```

Auto Commit | Manual

	lastName	firstName	jobTitle	officeCode
1	Patterson	Mary	VP Sales	1
2	Castillo	Pamela	Sales Rep	4
3	Gerard	Martin	Sales Rep	4
4	Nishi	Mami	Sales Rep	5
5	Bott	Larry	Sales Rep	7
6	Jones	Barry	Sales Rep	7

Dari gambar di atas dapat dilihat bahwa firstname semuanya berhuruf a pada huruf kedua dari namanya.

Selanjutnya kita masuk studi kasus lain, ingin menampilkan officeCode hanya 1, 3, 7. Kita bisa gunakan operator IN, logika ini sama dengan logika OR. Bila salah satu terpenuhi maka akan dianggap benar.

Contoh:

```sql
select * from employees;
select `lastName`, `firstName`, `jobTitle`, `officeCode`
from employees
where `officeCode` IN (1,3,7)
order by `officeCode`, `jobTitle`;
```

Auto Commit | Manual

	lastName	firstName	jobTitle	officeCode
1	Murphy	Diane	President	1
2	Bow	Anthony	Sales Ma...	1
3	Thompson	Leslie	Sales Rep	1
4	Jennings	Leslie	Sales Rep	1
5	Firrelli	Jeff	VP Mark...	1
6	Patterson	Mary	VP Sales	1
7	Tseng	Foon Yue	Sales Rep	3
8	Vanauf	George	Sales Rep	3
9	Bott	Larry	Sales Rep	7
10	Jones	Barry	Sales Rep	7

Kita juga bisa menampilkan untuk officeCode yang lebih besar dari 5. Caranya:

```
select * from employees;
select `lastName`, `firstName`, `jobTitle`, `officeCode`
from employees
where `officeCode` >5
order by `officeCode`, `jobTitle`;
```

Auto Commit | Manual

	lastName	firstName	jobTitle	officeCode
1	Patterson	William	Sales Ma...	6
2	Fixter	Andy	Sales Rep	6
3	Marsh	Peter	Sales Rep	6
4	King	Tom	Sales Rep	6
5	Bott	Larry	Sales Rep	7
6	Jones	Barry	Sales Rep	7

Studi kasus lain kita akan menampilkan kolom reportsTo:

```
select * from employees;
select `lastName`, `firstName`, `reportsTo`
from employees;
```

Auto Commit | Manual

	lastName	firstName	reportsTo
1	Murphy	Diane	(NULL)
2	Patterson	Mary	1002
3	Firrelli	Jeff	1002
4	Patterson	William	1056
5	Bondur	Gerard	1056
6	Bow	Anthony	1056
7	Jennings	Leslie	1143
8	Thompson	Leslie	1143

Bisa dilihat dari gambar di atas, bahwa Diane Murphy tidak memiliki reportsTo dengan ditandai sebagai NULL. NULL artinya kosong atau tidak ada. Maka, kalau kita ingin mencari reportsTo yang kosong bisa gunakan query seperti di bawah ini.

```sql
select `lastName`, `firstName`, `reportsTo`
from employees
where `reportsTo` is null;
```

Auto Commit | Manual

	lastName	firstName	reportsTo
1	Murphy	Diane	(NULL)

Contoh studi kasus 1:

```sql
///studi kasus 1: membuat query orderdetails dengan quantityOrdered di atas 40///
select * from orderdetails
where `quantityOrdered` >40;
```

Auto Commit | Manual

	orderNumber	productCode	quantityOrdered	priceEach	orderLineNumber
1	10100	S18_2248	50	55.09	2
2	10100	S24_3969	49	35.29	1
3	10101	S24_1937	45	32.53	3
4	10101	S24_2022	46	44.35	2
5	10102	S18_1367	41	43.13	1
6	10103	S10_4962	42	119.67	4
7	10103	S18_3320	46	86.31	16
8	10103	S18_4668	41	40.75	9
9	10103	S32_3522	45	63.35	7
10	10103	S700_2824	42	94.07	6
11	10104	S12_4473	41	111.39	9

Contoh studi kasus 2:

```sql
///studi kasus 2: membuat query dari table orders yang statusnya On Hold, Disputed, Cancelled///
select * from orders
where `status` in ('On Hold', 'Disputed', 'Cancelled');
```

Auto Commit | Manual | Save | Run Selection

	orderNumber	orderDate	requiredDate	shippedDate	status	comments
1	10167	2003-10-23	2003-10-30	(NULL)	Cancell...	Customer called to cancel. The warehouse was notified in time and the order didn't sh
2	10179	2003-11-11	2003-11-17	2003-11-13	Cancell...	Customer cancelled due to urgent budgeting issues. Must be cautious when dealing w
3	10248	2004-05-07	2004-05-14	(NULL)	Cancell...	Order was mistakenly placed. The warehouse noticed the lack of documentation.
4	10253	2004-06-01	2004-06-09	2004-06-02	Cancell...	Customer disputed the order and we agreed to cancel it. We must be more cautions w
5	10260	2004-06-16	2004-06-22	(NULL)	Cancell...	Customer heard complaints from their customers and called to cancel this order. Will r
6	10262	2004-06-24	2004-07-01	(NULL)	Cancell...	This customer found a better offer from one of our competitors. Will call back to renec
7	10334	2004-11-19	2004-11-28	(NULL)	On Hold	The outstaniding balance for this customer exceeds their credit limit. Order will be shi
8	10401	2005-04-03	2005-04-14	(NULL)	On Hold	Customer credit limit exceeded. Will ship when a payment is received.
9	10406	2005-04-15	2005-04-25	2005-04-21	Disput...	Customer claims container with shipment was damaged during shipping and some ite